Quantifying the cost of cultural bias in conservation decision making

1 Abstract

Human systems shape not only ecological data collection, but also perspectives and assumptions made during model building, generating uncertainty in how dominant value systems and cultural bias limit ecological inference. Understanding how this uncertainty propagates in conservation decision making requires translation beyond abstract statistical measures and toward performance on real-world objectives. Here we develop a quantitative framework combining uncertainty quantification and decision theory to understand the value of reducing uncertainty generated by cultural bias for a conservation decision maker. We demonstrate this framework in the context of decision making in the Columbia River Basin (CRB), where the Pacific lamprey – an ancient, foundational species in the CRB – modulates salmonid ocean survival both as a parasite of salmonids and a predation buffer against marine mammals. The Pacific lamprey, however, has faced a precipitous decline in the last century, largely stemming from Euro-American cultural bias, which has led to asymmetric measurement error and observation of local, rather than global, properties of the system. We show that multiple functional forms of this three-species dynamical system are statistically indistinguishable yet result in very different optimal management actions because of differences that lie outside the range of observed data. We use methods from decision theory to quantify the cost of lamprey uncertainty in units relevant for a decision maker, demonstrating that the value of information around overlooked species can be considerable. These results highlight how Situated Modeling in ecological management can be used to interrogate how modeling as a process and practice is contextualized.

2 Introduction

Building an ecological model is an active exercise of bringing about the world (Schlüter et al., 2025). Yet this world-making process is limited, as models tend to exist where the light is shining (Figure 1), often reflecting what is known and where the data is, rather than what is unknown. The presence or absence of parameters and processes in a model can reflect dominant systems of knowledge production, dominant societal values, and cultural bias (Amano & Sutherland, 2013; Silver et al., 2022; Stoudt et al., 2022). For decision makers, uncertainties that are known and articulated already challenge management of ecological systems (Polasky et al., 2011), yet understanding how cultural bias and systems of power generate uncertainties that propagate in conservation decisions remain largely unexplored.

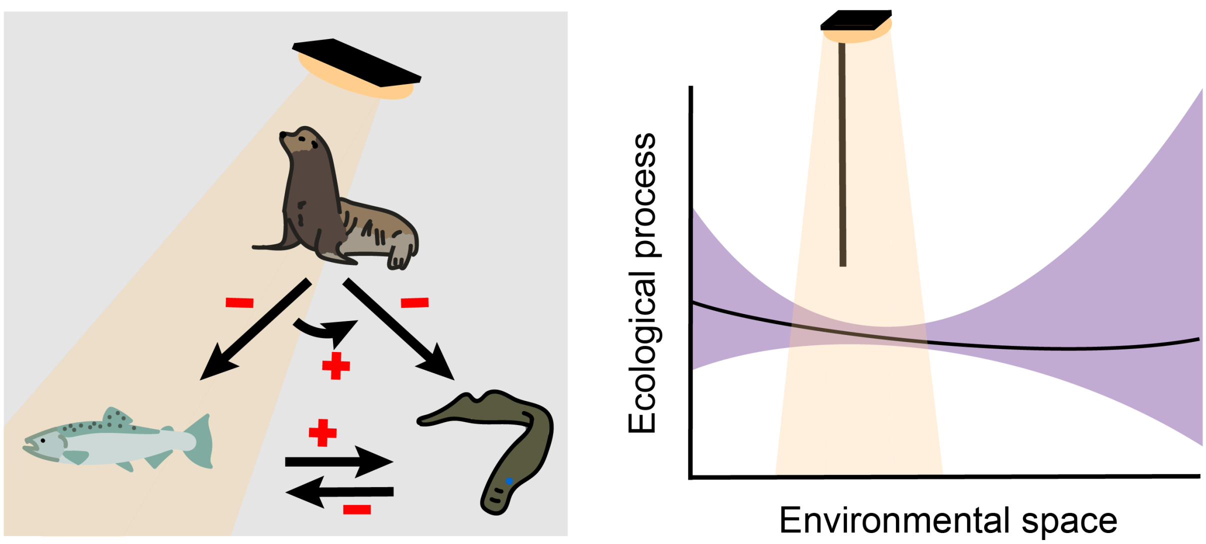


Figure 1: What is the cost of looking only under a lamp-post for a scientific solution, simply because that's where the light is? This focus as result of human bias can lead to unsampled, data-poor regions of environmental space, generating uncertainty in ecological processes that can propagate to uncertainty in the decision space. Here we present a methodological framework combining uncertainty quantification and decision theory that can be used to quantify the cost of cultural bias to a decision maker.

Windows of opportunity to observe complex ecological systems are not neutral, resulting in observations that can reflect human systems as much as they reflect the natural world. For example, global biodiversity observations emulate macro-economic spatial patterns (Amano & Sutherland, 2013), and human perceptions of species charisma shape citizen science reporting rates (Stoudt et al., 2022). This bias can generate uncertainty in ecological inference: observing only a subset of environmental space limits prediction of global system properties (Ellis-Soto et al., 2024) (Figure 1), and observation error can be uneven across species, locations, and points in time (Hughes et al., 2021). Translating the cost of this bias for decision makers requires moving beyond abstract statistical scores like confidence intervals and towards measurements on real-world objectives (Boettiger, 2022). In decision making, uncertainty matters when it affects the choice of management action (McCarthy & Possingham, 2007), and decision theory provides the tools to quantify the importance of uncertainty as the Value of Information (VoI), or the degree to which unresolved uncertainty affects performance on management objectives (Runge et al., 2011).

The Columbia River Basin (CRB) provides a fruitful setting to quantify the cost of cultural bias in conservation decision making (Box 1). Over the past few decades, the endangered, migratory Pacific salmon has faced increased predation pressure by California sea lions traveling into the Columbia River (Wargo Rub et al., 2019). This conflict between two federally protected and broadly beloved species has strained decision making around how to support both species, ultimately precipitating an amendment to the Marine Mammal Protection Act to allow legal euthanasia of these sea lions (Marine Mammal Protection Act § 120, 16 U.S.C. § 1389, 2018). Although the Columbia River salmon fisheries are among the most intensely managed and heavily studied fisheries in the world, one relatively understudied fish species – the Pacific lamprey – may play a role in restoring balance in this system (Figure 1). The Pacific lamprey holds high cultural and spiritual value for CRB Indigenous peoples and may relieve predation pressure on salmon as a high value prey item for marine mammals (Columbia River Inter-Tribal Fish Commission, 2011), yet the Pacific lamprey has been driven to near extinction over the last century, largely due to cultural bias and exclusion by Euro-American society and management (Box 1). The legacy of nearly a century of disregard is reflected in data about salmon-lamprey-sea lion interactions, as most data come from observation systems designed for salmon, not lamprey (Kostow, 2002), and understanding global, density-dependent dynamics is challenging, since lamprey abundance is much lower than historic levels (Figure 2; Figure S1).

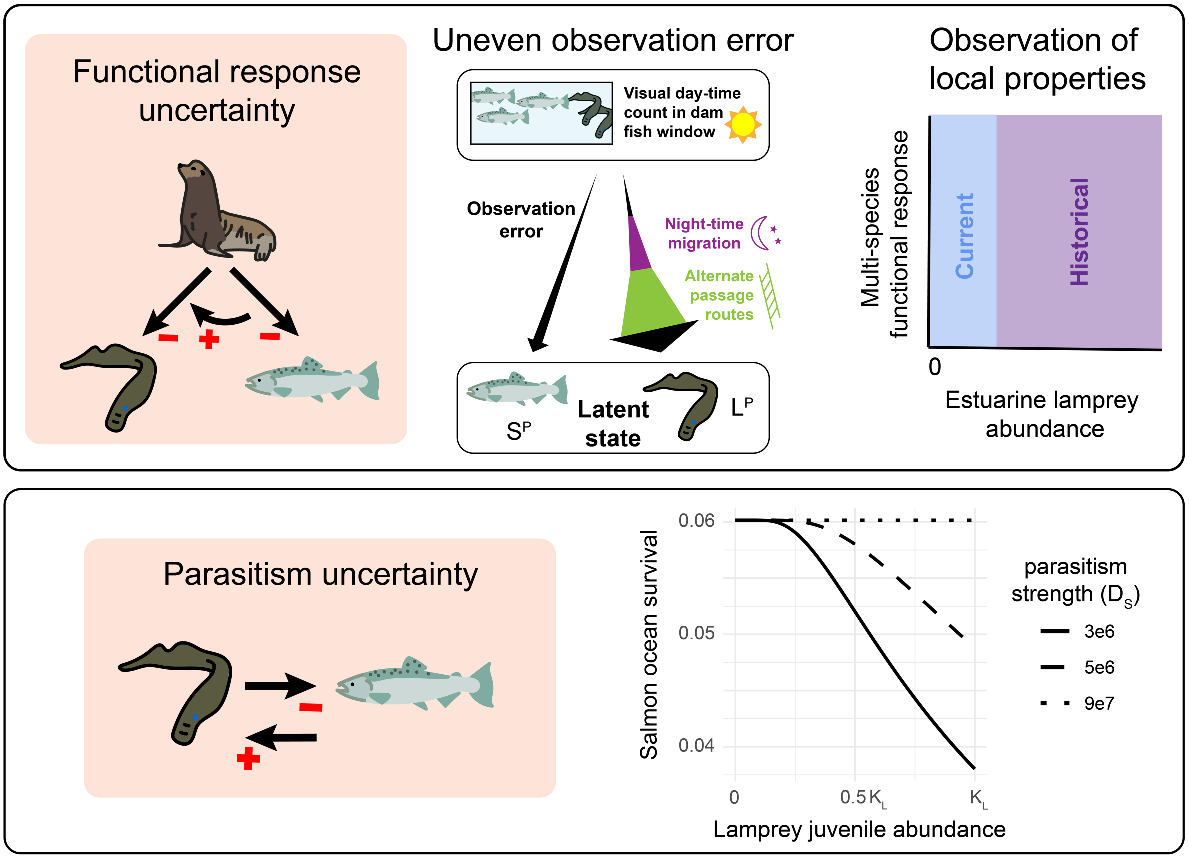


Figure 2: Two sources of parametric uncertainty considered in the decision problem: **A.** Functional response uncertainty, including the density-dependent rates of predation and prey switching, derived from 1) uneven observation error of the true abundance of fish passing through the dam (S^P and L^P for salmon and lamprey, respectively) and 2) observation of local properties of the density-dependent multi-species functional response (i.e., observation space limited to low Pacific lamprey estuarine abundance, E_L). **B.** Parasitism uncertainty, or an unknown rate at which salmon ocean survival (proportion of juvenile salmon returning the the estuary, E_S/J_S) declines with juvenile lamprey abundance (J_L). The parameter D_S describes this rate, and values are selected to cover the universe of possibilities, from a commensal lamprey-salmon marine relationship (D_S = 9e7), to a scenario where salmon ocean survival declines to 0.04 at high lamprey densities (D_S = 3e6).

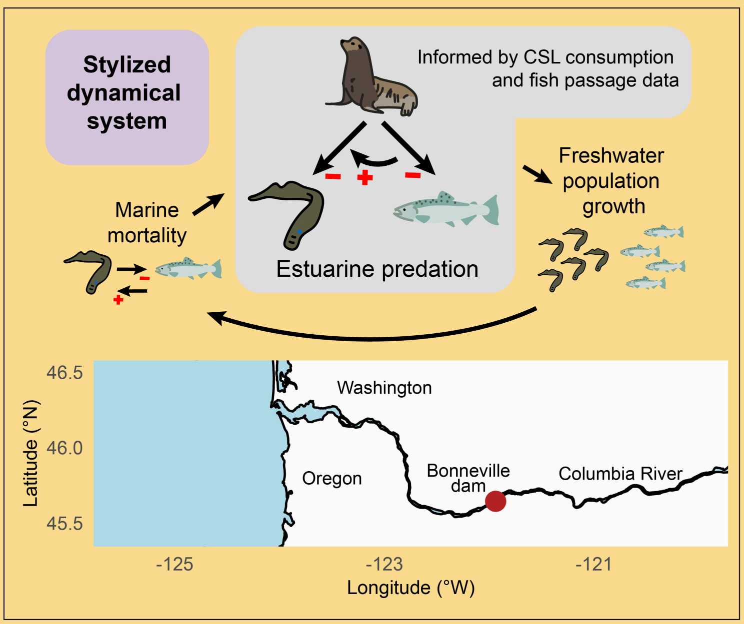
In this study, we quantify the extent to which historical cultural bias against the Pacific lamprey impedes decision making in the CRB. We first quantify uncertainty in the multi-species functional response (MSFR), or the functional form describing the rate of California sea lion predation as a function of the abundance of Pacific lamprey and spring Chinook salmon the Columbia River estuary (Rosenbaum et al., 2024). This model quantifies global uncertainty in the MSFR and is developed to consider how systems of knowledge production have generated asymmetries in the observation-generating processes between fish species (Carlen et al., 2024) (Figure 2). We then embed the quantified MSFR uncertainty into a decision model to quantify the value of resolving uncertainty related to the lamprey's role in modulating salmon survival as both a parasite and predation offset for salmon (Figures 1-2). This value of resolving uncertainty - Value of

Information - can be considered in the inverse, as the cost of a knowledge gap generated as an outcome of cultural bias. Through illustration with this Pacific lamprey case study, we provide a demonstrate a rigorous, quantitative approach to understanding the cost of cultural bias in conservation decision making, moving from legacies of disregard reflected in data to performance on decision objectives.

Box 1. Pacific lamprey, Chinook salmon, and California sea lions in the Columbia River Basin

Construction of hydroelectric dams along the Columbia River has dramatically altered flow regimes and fish passage, particularly for migratory Pacific salmonids of high cultural and economic value protected under the Endangered Species Act. While millions of dollars have been invested in freshwater habitat restoration and fish ladder construction, in the past few decades these fish have faced a new source of mortality: pinnipeds including California sea lions that have learned to travel to natural pinch points like dams and fish ladders to consume these fish by the thousands. The number of pinnipeds at these predation hotspots is increasing, due to recovery efforts related to the Marine Mammal Protection Act (MMPA) and ocean warming events that have shifted the geographic range of marine mammals in the California Current (Wargo Rub et al., 2019). To mitigate the effects of sea lion predation on salmon stocks at risk of extinction, the MMPA was amended in 2018 to authorize states and tribes to lethally remove sea lions, yet these controversial euthanasia practices have strained decision-making processes intended to maximize salmonid returns to the Columbia River Basin (CRB).

An underacknowledged piece of uncertainty that may impede decisions around policies protecting both salmon and sea lions is the role of Pacific lamprey in modulating salmonid survival and salmonid-sea lion interactions. The Pacific lamprey, an ancient, foundational species in the CRB, alongside which nearly all extant species in the CRB have co-evolved (Close, 2002). The Pacific lamprey serves as a predation buffer for upstream migrating adult salmon from marine mammals, as the species is extraordinarily rich in fats and relatively easy to capture (CRITFC, 2011; Roffe and Matte, 1984). The Pacific lamprey may also decrease salmonid survival in the marine environment, as the species plays a parasitic role in its adult stage, though the impact of parasitic attack wounds on salmonid fitness and survival is not well characterized (Beamish 1980; Weitkamp et al., 2015).



Despite its critical ecological role in the CRB and its high cultural value for Columbia basin Indigenous peoples (Close et al., 2002; 2004), the Pacific lamprey has faced a precipitous decline in the last century, stemming in part from Euro-American cultural bias and association with invasive lamprey in the Laurentian Great Lakes.

> "*[L]amprey are considered pests or trash fish by the dominant society, the result of which is the development of eradication programs or at best, a management policy of benign neglect*" (Close 2004).

While attention toward Pacific lamprey decline has increased in the last decade—largely due to advocacy, research, and knowledge generated by the Columbia River Inter-Tribal Fish Commission—the legacy of this bias persists in data. Since Pacific lamprey abundance is low relative to historic levels, understanding density-dependence in the lamprey-salmonid-sea lion functional response is difficult, as only local, rather than global, properties of this functional response are observed. Additionally, most data on lamprey are collected incidental to monitoring of salmonids, where the design and efficiency of the data collection effort is not always adequate for lampreys, resulting in amplified observation error for Pacific lamprey (Kostow, 2002; Tidwell et al., 2022).

3 Results

3.1 Multi-species dynamical system

We first built a stylized model of a Chinook salmon and Pacific lamprey dynamical system throughout their lifecycles as anadromous fish, including estuarine predation by California sea lions, freshwater population growth, and marine mortality (Box 1). We modeled Chinook salmon and Pacific lamprey interactions in 1) the marine environment through a density-dependent host-parasite relationship and 2) the estuarine environment through a density-dependent, multi-species functional response (MSFR), where the rate of California sea lion consumption is a function of the abundance of lamprey and salmon in the prey community. These two lamprey and salmon interactions were used to represent the rate at which lamprey modulate salmon survival as both a parasite of salmon and a predation offset for salmon. The parameter values used to describe freshwater population growth and marine mortality were guided by historical observations and are described in Table S1, whereas the MSFR parameters were directly estimated with data.

3.2 Quantified multi-species functional response uncertainty

To quantify uncertainty in the California sea lion multi-species functional response (MSFR) (Equations 1-2), we developed a Bayesian model that relates sea lion prey consumption data from gastro-intestinal diet analysis to the observed count of fish passing through Bonneville dam. The posterior distribution was used to quantify parametric uncertainty in the species-specific, density-dependent attack rates, b_i, species-specific handling times, h_i, and MSFR exponent, q in Equations 1-2 (Table S2, Figure S2).

The Bayesian model was used to estimate abundance of prey available to the predator below Bonneville dam by accounting for error in observing the abundance of fish passing through the dam and imperfect dam passage efficiency (Figure 3A, Figure S3, Table S2). Across most sampled time points, the abundance of Pacific lamprey was less than 10% of the abundance of Chinook salmon (Figure 3A). Yet despite this low observed abundance, Pacific lamprey were frequently over-represented in the sea lion's diet (Figure 3A).

The relatively low abundance of lamprey in the prey community limited observation of global properties of the multi-species functional response, creating high parametric uncertainty in model regions where lamprey make up more than 25% of the salmon-lamprey prey community (Figure 3B-C). This observation of local properties of the density-dependent MSFR prevented distinction between a global preference for lamprey, where lamprey are favored at all densities, and negative switching, where lamprey switch from being over-represented to under-represented in the diet of sea lions as lamprey density increases (Figure 3B-C). The model's posterior samples captured this global uncertainty (Figure 3C; Table S2; Figure S2).

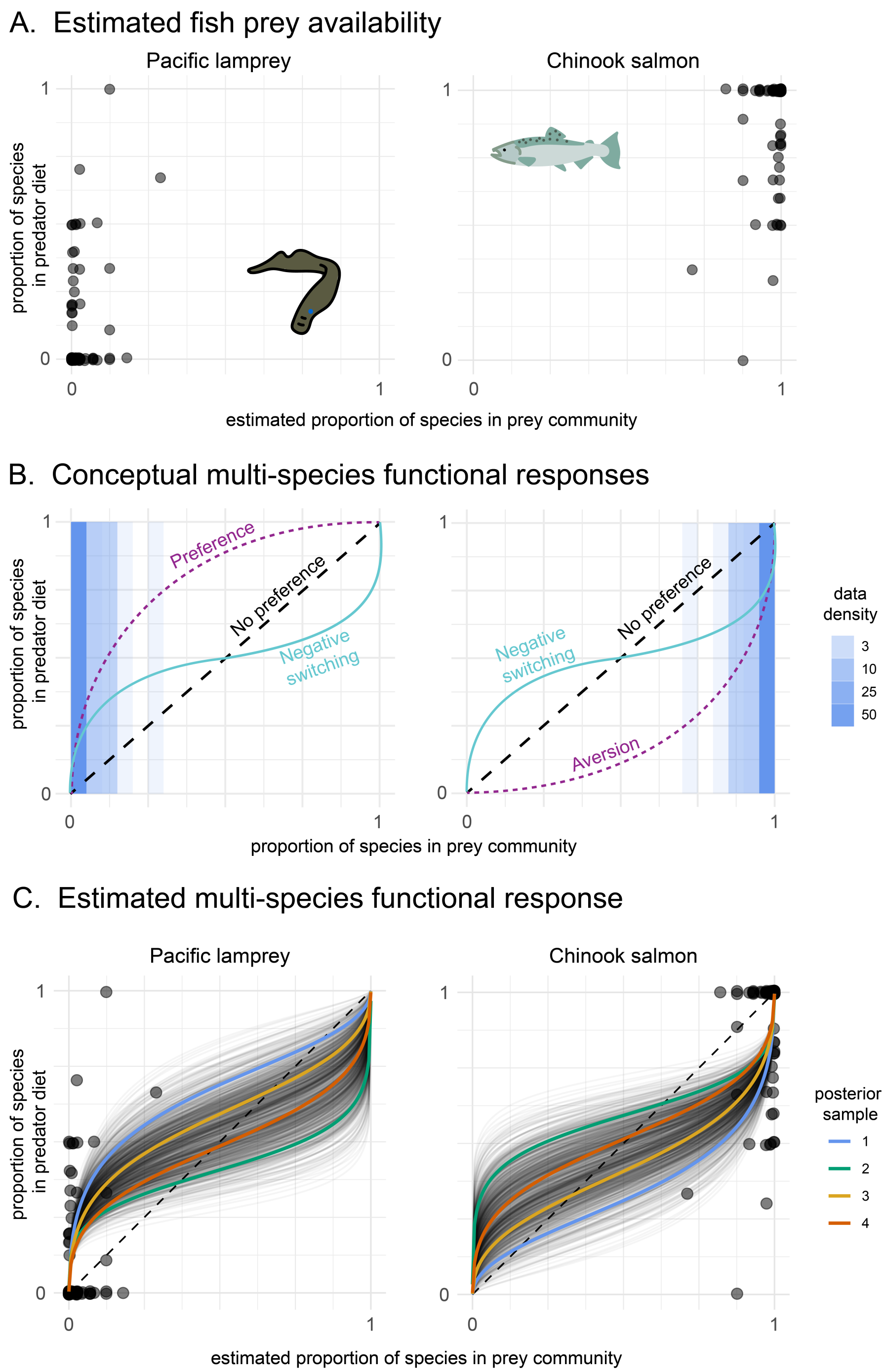


Figure 3: Multi-species functional response, measuring the rate at which California sea lions consume prey as a function of the composition of the prey community. **A.** Relationship between 1) the proportion of each fish species in the predator diet, measured with gastro-intestinal analysis of euthanized California sea lions, and 2) proportion of each fish species in the prey community, based on the mean posterior estimate of available prey, L_A and S_A for salmon and lamprey, respectively. The credibility intervals for the estimates of available prey, L_A and S_A, are shown in Figure S3. **B.** Conceptual multi-species functional responses demonstrating how two multi-species functional response curves can have similar local properties in the observed parametric region shown in panel A, yet different global properties due to a large, unobserved parametric region. **C.** Estimated multi-species functional response, where each black line represents a posterior sample describing the predicted relationship between the proportion of each species in the sea lion diet and the proportion of each species in the prey community. Four posterior samples are highlighted and are used in Figure 4. The dashed line indicates the 1:1 line, where sea lion consume prey in equal proportion to their abundance in the prey community.

3.3 Decision model

We built a decision model to quantify the cost of lamprey-related uncertainty to a decision maker. The Pacific lamprey has both inherent value and spiritual and cultural value to humans, including as a First Food for Columbia River Basin tribes (Close et al., 2002; Columbia River Inter-Tribal Fish Commission, 2011). However, for the purposes of this decision problem, we primarily considered the value of Pacific lamprey through its ecological role as a predation offset for salmonids to reflect a decision maker whose values align with the dominant value system that prioritizes Pacific salmon. We therefore framed the decision problem as finding the action, $a*$, that maximizes utility, U, where utility is defined as the equilibrium abundance of Chinook salmon returning to spawn in freshwater, $\hat{R}_S$.

In the decision problem, we considered a set of actions, $a \in A$, that represent manipulation of lamprey production, or the value of parameter α_L. Since this production parameter, α_L, can be understood as the product of adult freshwater survival and intrinsic rate of population growth, an action of increasing α_L could refer to increasing adult lamprey survival (e.g., improving dam passage) and/or increasing lamprey reproduction (e.g., artificial lamprey propagation).

While many uncertainties exist in this system, we considered two sources of parametric uncertainty (Figure 2): 1) uncertainty in the density-dependent California sea lion functional response, F, quantified above and 2) uncertainty in the strength of parasitism, P, or the rate at which salmon ocean survival declines as a function of lamprey abundance. We calculated the utility, $U(a, f, p)$, of each action, a, using the dynamical system model (Equations 1-7) under all expressions of functional response and parasitism uncertainty, F and P, respectively.

The optimal action, $a*$, that maximized utility (i.e., equilibrium abundance of salmon returning to the Columbia River basin) varied across different expressions of functional response and parasitism uncertainty (Figure 4A; Figure S4). The utility associated with the optimal action, $\max_a U(a, f, p)$, also varied across different expressions of functional response and parasitism uncertainty (Figure 4B; Figure S4). In general, expressions of MSFR uncertainty associated with a global preference for lamprey (Figure 3C, blue line) corresponded to higher maximized expected utility, $\max_a E_P[U(a, f, p)]$ (Figure 4A, blue line). Whereas expressions of MSFR uncertainty associated with negative switching (Figure 3C, green line) corresponded to lower maximized expected utility (Figure 4A, green line; Figure S5). Additionally, the bet-hedging strategy, or the action that maximizes utility over all uncertainty ($\max_a E_{F,P}[U(a, f, p)]$), tended to be higher than the action that maximized utility after functional response uncertainty had been resolved ($\max_a E_P[U(a, f, p)]$) (Figure 4A; Figure S5).

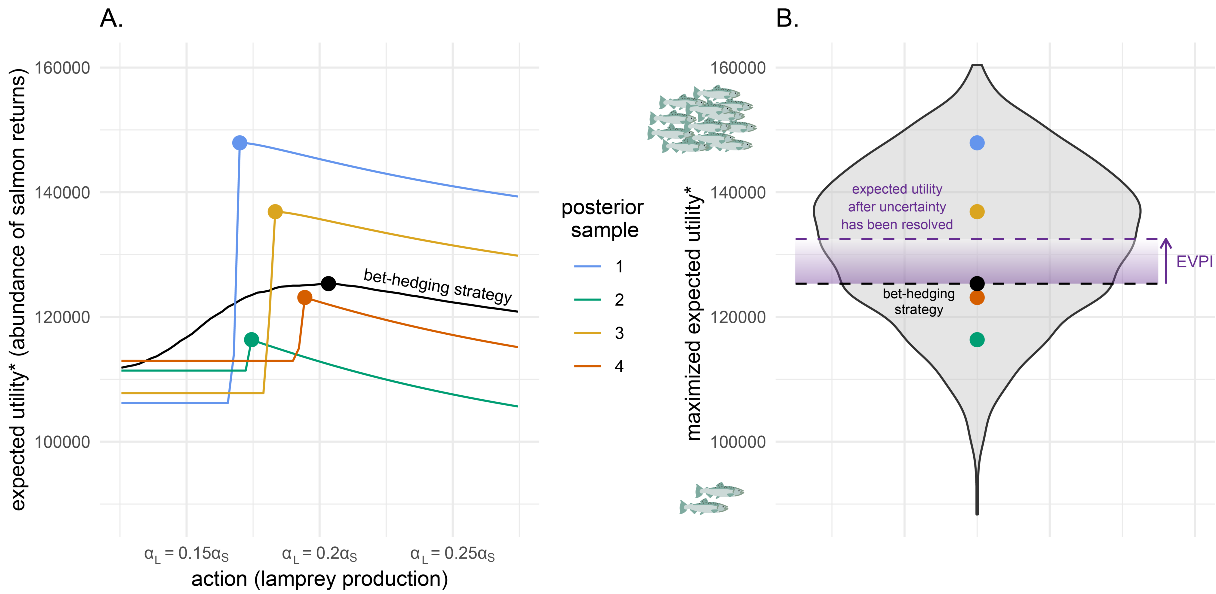


Figure 4: Quantifying the value of reducing lamprey-related uncertainty. **A.** Relationship between the expected utility and action for each multi-species functional response (MSFR) posterior sample highlighted in Figure 3C. Asterisk indicates that the expected utility, $E_P[U(a, f, p)]$, is calculated as the expectation over parasitism uncertainty, P (Figure 2). The actions are the lamprey production, α_L, relative to salmon production, α_S, and can be understood as actions that either increase freshwater lamprey adult survival or the intrinsic rate of lamprey population growth. The points indicate the action that maximizes expected utility (i.e., $\max_a E_P[U(a, f, p)]$) for each expression of functional response uncertainty. The black line indicates the expected utility over both parasitism and functional response uncertainty (i.e., $E_{F,P}[U(a, f, p)]$), and the black point indicates the action that maximizes utility over all uncertainty, or the bet-hedging strategy (i.e., $\max_a E_{F,P}[U(a, f, p)]$). **B.** Violin plot of the maximized expected utility of all posterior samples, calculated as the expectation across parasitism uncertainty (i.e., $\max_a E_P[U(a, f, p)]$). Colored points indicate the maximized expected utility of the selected posterior samples from panel A. The black point indicates the maximized expected utility across both parasitism and functional response uncertainty, or the utility associated with the bet-hedging strategy, $\max_a E_{F,P}[U(a, f, p)]$. The purple dashed line indicates the expected utility after all uncertainty has been resolved, $E_{F,P}[\max_a U(a, f, p)]$. The expected value of functional response and parasitism information (i.e., expected value of perfect information, EVPI) is therefore the difference between $E_{F,P}[\max_a U(a, f, p)]$ and $\max_a E_{F,P}[U(a, f, p)]$.

3.4 Cost of lamprey uncertainty to a decision maker

In the context of this decision problem, the value of lamprey information was quantified as the expected improvement on the management objective (maximize abundance of Chinook salmon returning to the Columbia River Basin) if functional response uncertainty, F, and parasitism uncertainty, P, were reduced (Runge et al., 2011). The utility associated with the bet-hedging strategy ($\max_a E_{F,P}[U(a, f, p)]$) was 125,343 fish, whereas the expected utility after all uncertainty had been resolved was 132,511 fish (Figure 4B). The expected value of perfect information, EVPI, was therefore 7167 fish (Figure 4B). Framed another way, the expected increase in Chinook salmon returning to the Columbia River Basin was 7167 fish, if lamprey uncertainty - generated as an outcome of cultural bias - was not impeding the choice of action.

4 Discussion

Evaluating the outcomes of conservation actions can be limited by model uncertainty that arises from cultural bias. To explore this challenge, we develop a decision model representing the dynamics and management of three interacting species in the Columbia River Basin, a useful setting for quantifying the extent to which dominant value systems impede conservation decision making. Here, management has historically focused on high-value salmon fisheries and recent predation pressure by sea lions, while overlooking a culturally and ecologically important species - the Pacific lamprey (Figure 1). This focus and historical bias against Pacific lamprey by dominant society has led to asymmetric measurement error and observation limited to local properties of the system with lamprey at low relative abundance (Figure 2, Figure 3).

An important implication of this historical bias is that multiple functional forms for the rate of sea lion predation on these two fish species fit the observed data equally well (Figure 3C), yet differ in resultant management policies (Figure 4A) (Runge & Johnson, 2002). These differing functional forms represent realistic density-dependent dynamics (Hellström et al., 2014; Rindorf et al., 2006), yet are statistically indistinguishable due to differences beyond the region of observed data (Figure 3B). To make this statistical uncertainty decision-relevant, we use a Value of Information analysis to quantify how additional information may improve management decisions. We show that resolving uncertainty related to the rates of California sea lion predation and Pacific lamprey parasitism would increase the expected number of Pacific salmon returning to the Columbia River by 5.7%. This increase in expected decision utility is significant, as the value of information is measured as the expected increase in decision performance relative to the bet-hedging strategy (Figure 4B), an action designed to perform optimally despite uncertainty (Holden et al., 2024). While the ecological community broadly acknowledges the ways in which humans and systems of power shape knowledge disparities and assumptions in ecological modeling (Chapman et al., 2024; Silver et al., 2022), this work makes an important step towards quantifying how legacies of cultural bias in environmental management impede today's decisions.

4.1 Reconciling unknown unknowns

Quantifying the cost of cultural bias to a decision maker through a Value of Information analysis is only possible if a spark brings to light previously ignored sources of uncertainty. In the case of the Pacific lamprey, Native American tribes began championing the importance of the species in the early 1990s, recognizing the declining numbers of lamprey and reflecting on the role of dominant Euro-American societal values in contributing to their decline (Clemens et al., 2017; Close et al., 2002, 2004). There is, however, not a VOI calculus for uncertainty that is not articulated, reflecting the philosophical conundrum of understanding the cost of "unknown unknowns" in scenarios where the truth lies far outside of the articulated hypotheses (Wintle et al., 2010).

However, structural uncertainty in ecological modeling is often not random, as science is not neutral to power and uneven knowledge can arise systematically from political-economic structures and processes (Silver et al., 2022). Moving from "unknown unknowns" to quantifying the cost of "known unknowns" can involve interrogating systems of knowledge production through long-established ideas from the fields of environmental justice and science and technology studies. Investigating how environmental and social processes are intertwined can provide valuable insight into how uneven distribution of power shapes conservation investment and research (Walker, 2012). Considering knowledge as situated and knowing is a social process can be crucial for reflecting upon the systems of scientific practice (Bloor, 2004; Haraway, 2013). For instance, reflection on the relationship between a perceived ecological role and necropolitics – who should live and who should die – can be useful for interrogating the relationship between human psychology and conservation policy (Chao, 2021). Integrating these ways of thinking into statistical ecology and decision making will be crucial for understanding how legacies of human structures can be present or create absences in ecological model development.

4.2 Implications for conservation decision making

Conservation decision making is always bias- and value-laden; in fact, decision theory encourages value-based thinking and posits that values should be the driving force for decision making (Keeney, 1996). Danger, however, comes from a lack of reflection on where values enter the decision-making process. Decision analysis traditionally assumes that facts and values are separable: values are used to formulate the decision context and objectives, and facts are used to determine the consequences of actions (Gregory et al., 2012). Here we have shown, however, that historical values of a dominant society affect the ability to predict the consequences of actions, generating a quantifiable impediment to decision making about salmon and lamprey in the Columbia River Basin.

While the inseparability of facts and values may challenge the conceptual foundation of decision analysis and structured decision making, including reflexive and relational thinking in adaptive management cycles represent a productive path forward. Adaptive management (AM) involves acknowledging uncertainty and seeking to reduce it through the process of management; AM can include two learning cycles, a technical learning cycle nested within a larger cycle of learning about the decision structure itself (Williams & Brown, 2016). To reduce uncertainty in how systems of power shape knowledge production and impede decision making, this outer loop of double-loop learning could involve processes by which a decision analyst extends a relational understanding of the world to modelling, a practice known as "Situated Modelling" (Klein et al., 2024). This practice includes reflecting on the broader context in which knowledge is generated, questioning where modeling assumptions come from and what is left out, and broadening perspectives and ways of knowing at the decision-making table.

4.3 Limitations of an utilitarian understanding of information value

In this analysis, we chose to evaluate the cost of lamprey uncertainty with respect to one decision objective - maximize the abundance of Chinook salmon returning to the CRB - among many possible decision objectives. This deliberate choice to frame the value of lamprey through the lens of its ecosystem service for salmon was chosen to represent how a historic, dominant value system that prioritized Pacific salmon can impede decision making, even if the value system remains unchanged.

The cost of cultural bias in decision making, however, would change with a different decision objective or a different understanding of cost. While we translated signatures of cultural bias reflected in ecological data to a measurement of cost in units relevant to a decision maker (i.e., expected number of salmon), this utilitarian representation reflects a narrow understanding of information value. Cultural bias-derived uncertainty can impede decision making in other ways, including generating conflict and diminishing trust among decision makers, thereby inhibiting the adaptive governance of the social-ecological system (Folke et al., 2005).

4.4 Conclusion

By combining uncertainty quantification with decision theory, we have demonstrated how to measure the cost of overlooking the Pacific lamprey in decision making in an important regional fishery. The Bayesian approach to uncertainty quantification offers a flexible framework for representing uneven data-generating processes, and the Value of Information analysis facilitates systematic exploration of this uncertainty in a decision-making context. By unifying these approaches, we quantify the extent to which uncertainty about Pacific lamprey impedes decision making about Chinook salmon, which can be understood as the cost of cultural bias, measured in units relevant to a decision maker.

5 Materials and Methods

We first describe a stylized representation of a Chinook salmon and Pacific lamprey dynamical system (Box 1). We then quantify uncertainty in estuarine predation using a Bayesian multi-species functional response (MSFR) model using sea lion gastro-intestinal diet data and fish passage data at Bonneville dam. Finally, we build a decision model that accounts for multiple sources of uncertainty in the dynamical system to evaluate management actions with respect to an objective of maximizing Chinook salmon equilibrium abundance. Using a Value of Information (VOI) analysis, we quantify how parametric uncertainty - generated as an outcome of cultural bias - impedes decision making.

5.1 Multi-species dynamical system

Chinook salmon and Pacific lamprey are anadromous fish, migrating from freshwater rivers to the ocean and back to spawn in or near their natal streams (Hess et al., 2023). While their lifecycles in the freshwater and marine systems extend across many years, we simplify the cycle to occur across one time step from t to $t+1$.

5.1.1 Estuary predation

The expected number of Pacific lamprey and Chinook salmon consumed by California sea lions is described using a multi-species functional response (MSFR) that allows the attack rate to be dependent on prey density (Rosenbaum et al., 2024). Here, $\widetilde{F}_i$ represents the expected number of prey i consumed per predator integrated across one day, a_i represents the density-dependent attack rate of species i, h_k represents the handling time for species $k = 1 \ldots m$, and E_i represents the abundance of species i entering the estuary.

$$\widetilde{F}_i = \frac{a_i E_i}{1 + \sum_{k=1}^{m} a_k h_k E_k} \tag{1}$$

The density-dependent attack rate, a_i, is a function of coefficient, b_i, and exponent, q. When $q = 0$, the MSFR follows a Holling type 2 functional response, and when $q < 0$, the MSFR follows a Holling type 3 functional response (Figure S6).

$$a_i = b_i E_i^q \tag{2}$$

The relationship between the abundance of fish returning to freshwater to spawn, R_i, and adults entering the estuary, E_i, of species $i \in \{L, S\}$ is therefore a function of the expected number of prey, $\widetilde{F}_i$, and the expected number of sea lion days (i.e., daily abundance in estuary $\times$ days), P:

$$R_i = E_i - \widetilde{F}_i \times P \tag{3}$$

5.1.2 Freshwater population growth

The relationship between 1) the abundance of fish returning to the Columbia River Basin to spawn, R_i and 2) the abundance of juveniles/smolts exiting Bonneville dam downstream, J_i, is described with the Ricker stock-recruit relationship, where α is a production parameter, and K is the smolt or juvenile carrying capacity. The production parameter, α, combines both freshwater mortality and the population intrinsic rate of growth.

We use the following growth equations to describe the relationship between the abundance of returning fish, R_i, and abundance of juveniles, J_i, for fish species i:

$$J_i = R_i \times \frac{\alpha_i}{(1 + \beta_i R_i)} \tag{4}$$

$$\beta_i = \frac{\alpha_i}{K_i} \tag{5}$$

5.1.3 Marine mortality

In the marine environment, Pacific lamprey and Chinook salmon exhibit a host-parasite relationship (Clemens et al., 2017). Salmonid host abundance in the marine environment is a principal factor in predicting Pacific lamprey returns to the Columbia River (Murauskas et al., 2013), and while Chinook salmon are common lamprey hosts and exhibit wounds from Pacific lamprey bites (Shevlyakov & Parensky, 2010; Weitkamp et al., 2015), the effect of these bites on their physiological condition is not well understood (Pelenev et al., 2008).

The relationship between the abundance of lamprey juveniles entering the ocean, J_L, and the abundance of lamprey adults returning to the estuary, E_L, is described by the following equation, where survival increases as a function of salmon density, J_S, at a rate described by D_L:

$$E_L = J_L \times s^o \times (1 - \mathrm{e}^{-D_L \times J_S}) \tag{6}$$

The relationship between the abundance of salmon smolts entering the ocean, J_S, and the abundance of salmon adults returning to the estuary, E_S, is described by the following equation, where survival decreases as a function of lamprey density, J_L, at a rate scaled by D_S:

$$E_S = J_S \times s^o \times (1 - \mathrm{e}^{-D_S/J_L}) \tag{7}$$

Here, density-independent ocean survival, s^o, is shared between both species.

5.2 Quantifying uncertainty in the multi-species functional response (MSFR)

Below we outline the Bayesian model formulation used to estimate the multi-species functional response (MSFR) parameters with California sea lion gastro-intestinal diet data and fish count data at Bonneville dam (Tables S3-S6). More information on data sources and model fitting can be found in Supporting Information.

5.2.1 Chinook salmon prey abundance

The observed number of Chinook salmon passing through the Bonneville dam fish ladders in time window, t, S_t^P, is drawn from a Poisson distribution with $\widetilde{S_t^P}$ representing the expected number of salmon passed (Table S4):

$$S_t^P \sim \text{Poisson}(\widetilde{S_t^P}) \tag{8}$$

The number of salmon available for sea lion consumption, S_t^A is greater than the number of salmon passing through the dam, as the dam passage efficiency is less than one. The expected number of salmon passed, $\widetilde{S_t^P}$, is therefore drawn from a Binomial distribution with the size as the difference between the number of salmon available for consumption, S_t^A, and the total number of salmon consumed across all euthanized sea lion individuals at time t, $\sum_{j=1}^{O_t} F_{t,j}^S$. Here, O_t is the total number of individuals euthanized at time t, and $F_{t,j}^S$ is the number of Chinook salmon enumerated in the digestive tract of sea lion individual, j. The probability is the salmon passage efficiency at Bonneville dam, p_S, (Frick et al., 2008):

$$\widetilde{S_t^P} \sim \text{Binomial}(S_t^A - \sum_{j=1}^{O_t} F_{t,j}^S, p_S) \tag{9}$$

5.2.2 Pacific lamprey prey abundance

In the years 2017 and 2019 when the lamprey night-time and LPS counts were recorded, the observed number of lamprey passing through all passage structures (LPS + fish ladders) in time window t, L_t^P, is drawn from a Poisson distribution with $\widetilde{L_t^P}$ representing the expected number of lamprey passed (Table S4):

$$L_t^P \sim \text{Poisson}(\widetilde{L_t^P}) \qquad \text{for } t \in \lambda^E \tag{10}$$

where λ^E represents the set of time windows associated with sea lion euthanasia in 2017 or 2019.

In the years when the lamprey night-time and LPS counts were not recorded, the observed number of lamprey passing through all passage structures, L_t^P, is drawn from a Binomial distribution, with the size parameter as the expected number of lamprey passed, $\widetilde{L_t^P}$, and probability of detection in daytime visual counts, p_t^D (Table S4):

$$L_t^P \sim \text{Binomial}(\widetilde{L_t^P}, p_t^D) \qquad \text{for } t \in \lambda^I \tag{11}$$

where λ^I represents the set of time windows associated with sea lion euthanasia in 2018 or 2021-2023.

The probability of detection in daytime visual counts during the time window t, p_t^D, is drawn from a Beta distribution:

$$p_t^D \sim \text{Beta}(\alpha^D, \beta^D) \qquad \text{for } t \in \lambda^I \tag{12}$$

These beta distribution hyperparameters were informed by the passed lamprey recorded via visual count, C_d^V, and the total recorded passed lamprey across all passage structures and all times of day, C_d^T, for each monitored three-day interval, d, in 2017 and 2019 (Table S5, Figure S7). C_d^V/C_d^T therefore represents the fraction of all passed lamprey that were visually counted during the day:

$$C_d^V/C_d^T \sim \text{Beta}(\alpha^D, \beta^D) \tag{13}$$

The number of lamprey available for sea lion consumption, L_t^A is greater than the number of lamprey passing through the dam, as the dam passage efficiency is less than one. The expected number of lamprey passed, $\widetilde{L_t^P}$, is drawn from a Binomial distribution with the size as the difference between the number of lamprey available for consumption, L_t^A, and the number of lamprey consumed across all euthanized sea lion individuals at time t, $\sum_{j=1}^{O_t} F_{t,j}^L$. Here, $F_{t,j}^L$ is the number of lamprey enumerated in the digestive tract of sea lion individual, j. The probability is the lamprey passage efficiency at Bonneville dam, p_L, (Moser et al., 2002):

$$\widetilde{L_t^P} \sim \text{Binomial}(L_t^A - \sum_{j=1}^{O_t} F_{t,j}^L, p_L) \tag{14}$$

5.2.3 Consumed prey

The number of prey consumed by each sea lion individual is likely to deviate from the expected number of prey consumed due to heterogeneity in preference and behavior across sea lion individuals. Model process error is therefore incorporated as Poisson-distributed variation in sea lion consumption rates. The observed number of Chinook salmon and Pacific lamprey consumed by sea lion individual j, $F^S_{t,j}$ and $F^L_{t,j}$, respectively, is drawn from a Poisson distribution with the expected number of prey consumed, $\widetilde{F^S_t}$ and $\widetilde{F^L_t}$, as the rate parameters, respectively (Table S6):

$$F^S_{t,j} \sim \text{Poisson}(\widetilde{F^S_t}) \tag{15}$$

$$F^L_{t,j} \sim \text{Poisson}(\widetilde{F^L_t}) \tag{16}$$

5.2.4 Multi-species functional response (MSFR)

The expected number of prey consumed, $\widetilde{F_{i,t}}$, of species i is related to the available number of prey, $A_{i,t} = \{S^A_t, L^A_t\}$, using the MSFR described in Equations 1-2.

$$\widetilde{F_{i,t}} = \frac{a_{i,t}A_{i,t}}{1 + \sum_{k=1}^{m} a_{k,t}h_k A_{k,t}} \tag{17}$$

$$a_{i,t} = b_i A^q_{i,t} \tag{18}$$

5.3 Decision model

We then developed a decision model, where we framed the decision problem as finding the action, $a*$, that maximizes utility, U, where utility is defined as the equilibrium abundance of Chinook salmon returning to spawn in freshwater, $\hat{R}_S$. The set of actions, $a \in A$, under consideration in the decision problem include changes in lamprey production, α_L. While many uncertainties exist in this system, we considered two sources of parametric uncertainty (Figure 2): 1) functional response uncertainty, F (Figure 3) and 2) parasitism uncertainty, P, describing uncertainty in the rate at which salmon ocean survival declines as a function of lamprey abundance.

We calculated the utility of each action, a, using the dynamical system model (Equations 1-7) under all combinations of functional response and parasitism uncertainty, F and P. Here, $U(a, f, p)$ is the utility associated with taking action a assuming $f \in F$ and $p \in P$. The set F corresponds to 1000 randomly selected samples from the MSFR model posterior distribution, and the set P corresponds to three values of

D_S (Equation 7) with equal prior probability (Figure 2, Table S1). The utility, or the equilibrium salmon return abundance ($\hat{R}_S$), was calculated through simulation, starting at an arbitrary abundance of salmon and lamprey returns, R_S and R_L, and finding the mean salmon return abundance, $\hat{R}_S$, after removing transient dynamics.

5.3.1 Value of Information

We then used a Value of Information analysis to calculate the value of lamprey information. In classical decision theory, the expected value of perfect information (EVPI), is the difference between the expected value of an optimal action after new information has been collected and the expected value of an optimal action before new information has been collected:

$$\text{EVPI} = E_{F,P}[\max_a U(a,f,p)] - \max_a E_{F,P}[U(a,f,p)] \tag{19}$$

The first term in Equation 19 represents the expected utility once all uncertainty has been resolved, because the optimal action is chosen after perfectly knowing the multi-species functional response relationship and how parasitism strength scales with lamprey density. The second term in Equation 19 is the bet-hedging strategy, or the expected value associated with the action that maximizes utility over all sources of uncertainty. The difference between these two terms, EVPI, is therefore defined as the expected increase in utility if lamprey uncertainty is resolved.

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